

Instagram Likes Prediction

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I used regression to predict how many likes an Instagram post would receive. The best model in the final comparison combined metadata with nine visual clusters formed from CLIP features and a Random Forest. It achieved a test mean absolute error (MAE) of 67,675 likes, reducing the average error by 45.7% compared with the median baseline.

Data and evaluation

The dataset contains 3,785 posts with images, likes, comments, follower counts and timestamps. Every post matched an image file and the required fields had no missing or invalid values, so all the rows were kept as it is. Eight extra copies of identical images, but none appeared across different data splits. I sorted the posts by time and used the oldest 2,271 posts (60%) for training, the next 757 (20%) for validation and the latest 757 (20%) for testing. The models were selected using validation MAE, then refitted on training and validation data before testing.

Approach

The training plots showed that most posts had relatively modest like counts, while a few had much higher counts. Followers and likes had a weak rank correlation (Spearman 0.147). I started with a median baseline, then tried linear regression, log transformations, Ridge and Random Forest. I created eight metadata features using log follower count, sine and cosine representations of UTC posting hour, weekday and month, and elapsed time. I excluded comments and filenames from the inputs.

For the images, pretrained CLIP ViT-B/32 was used to extract 512 normalized features per image and kept its weights frozen. KMeans grouped the training images into nine visual clusters without using their likes. I represented each cluster with a one-hot indicator and combined these nine indicators with the eight metadata features. This gave 17 inputs to a Random Forest with 200 trees and a minimum leaf size of 5. The model predicted $\log(1 + \text{likes})$, which I converted back into like counts. I used nine clusters based on my earlier experiments.

Results on 757 Test data

Model	Test mean absolute error in likes
Metadata + nine CLIP clusters + Random Forest	67,675
Metadata + Random Forest	76,064
CLIP features + Ridge	97,502
Metadata + full CLIP features + Random Forest	104,358
Median baseline	124,613

Main findings

Adding visual clusters reduced MAE by 11.0% compared with using metadata alone, with an R^2 of 0.656. Using all 512 CLIP features did not improve the models I tested, but grouping similar images helped. In one selected example, post 8192.jpg received 107,815 likes, while the model predicted 114,786 which is a difference of 6,971 likes. When I converted the predictions into low, medium and high like bands, the model achieved 64.7% band accuracy and a macro F1 score of 0.622. The quarter of test posts with the most likes had the largest average absolute error at 140,318 likes.

Limitations and next steps

I used one chronological validation split. Next, I would use time-series cross-validation to check model performance across different periods. I also noticed that posts with very high like counts were often underpredicted. The dataset does not include content creator account IDs or specify how long likes were counted after posting. We also need to confirm when follower counts were recorded. Since I checked test results while trying different approaches, I would evaluate the final model on new, later posts to confirm its performance.